

2	(a)	(i)	$W = p \Delta V = 57.0 \times 10^5 \text{ Pa} \times (3.1 - 2.0) \times 10^{-5} \text{ m}^3$ $= 62.7 \text{ J}$																								
		(ii)	$W = 0 \text{ J}$																								
	(b)		<table><tr><th>Stage of cycle</th><th>heat supplied to gas /J</th><th>work done on gas /J</th><th>increase in internal energy of the system /J</th></tr><tr><td>$P \rightarrow Q$</td><td>0</td><td>235</td><td>$\Delta U = Q + W$ $= 0 + 235 =$ 235 [A]</td></tr><tr><td>$Q \rightarrow R$</td><td>246</td><td>-62.7 [B] Since it is work done ON gas, put a negative sign.</td><td>$\Delta U = Q + W$ $= 246 - 62.7 =$ 183 [A]</td></tr><tr><td>$R \rightarrow S$</td><td>0</td><td>-333</td><td>$\Delta U = Q + W$ $= 0 - 333$ $= -333 \text{ [A]}$</td></tr><tr><td>$S \rightarrow P$</td><td>$\Delta U = Q + W$ $-85 = Q$ $Q = -85 \text{ [A]}$</td><td>No change in vol $\rightarrow$ no work done 0 [C]</td><td>$\Delta U = 0$ $235 + 183 - 333 +$ $\Delta U_{S \rightarrow P} = 0$ $\Delta U_{S \rightarrow P} = -85 \text{ [D]}$</td></tr></table> [A] (application of first law equation – mark by merit), [B] (insertion of negative sign), [C] (identifying work done is 0), [D] ($\Delta U = 0$)				Stage of cycle	heat supplied to gas /J	work done on gas /J	increase in internal energy of the system /J	$P \rightarrow Q$	0	235	$\Delta U = Q + W$ $= 0 + 235 =$ 235 [A]	$Q \rightarrow R$	246	-62.7 [B] Since it is work done ON gas, put a negative sign.	$\Delta U = Q + W$ $= 246 - 62.7 =$ 183 [A]	$R \rightarrow S$	0	-333	$\Delta U = Q + W$ $= 0 - 333$ $= -333 \text{ [A]}$	$S \rightarrow P$	$\Delta U = Q + W$ $-85 = Q$ $Q = -85 \text{ [A]}$	No change in vol $\rightarrow$ no work done 0 [C]	$\Delta U = 0$ $235 + 183 - 333 +$ $\Delta U_{S \rightarrow P} = 0$ $\Delta U_{S \rightarrow P} = -85 \text{ [D]}$	
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	(c)		$\frac{P_P V_P}{T_P} = \frac{P_Q V_Q}{T_Q}$ $T_Q = \frac{P_Q V_Q T_P}{P_P V_P} = \frac{(57.0 \times 10^5)(2.0 \times 10^{-5})(300)}{(1.00 \times 10^5)(36.0 \times 10^{-5})}$ $= 950 \text{ K}$																								

3	(a)	The graph is a straight line passing through the origin which shows that the acceleration is	
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